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Roll No

IT-504 (GS)
B.E. V Semester
Examination, June 2020
Grading System (GS)
System Programming and Operating System
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions
ii) All questions carry equal marks.

1. Describe the differences between symmetric and asymmetric multiprocessing. What are three advantages and one disadvantage of multiprocessor system?

2. Consider the following set of processes:

Process	Arrival time	Service Time
P ₁	0	3
P ₂	2	6
P ₃	4	4
P ₄	6	5
P ₅	8	2

Calculate response time, average waiting time and turnaround time. For FCFS, SJF (Preemptive and Non-Preemptive), Round Robin (Time Quantum=3unit)

3. Explain the following term :

- i) Compiler
- ii) Macros
- iii) Linkers
- iv) Assemblers

4. Define following terms:

- i) System Software
- ii) Semantic Gap
- iii) Specification Gap
- iv) Execution Gap

5. What is multiprogramming? State objective of multiprogramming.

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6. A Disk Drive having 5000 cylinders 0 to 4999. The Drive is currently serving a request at cylinder 143, and the previous request was at cylinder 125. The queue of pending requests, in FIFO order is :

86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130

Compute total head movements using following algorithms :

- i) FCFS
 - ii) SSTF
 - iii) SCAN
 - iv) LOOK
7. Write a short notes on:
- i) Thrashing
 - ii) Paging
 - iii) Cache memory
 - iv) Virtual memory
8. Write a short notes on (any three) :
- a) Distributed system
 - b) PCB
 - c) Demand paging
 - d) Software tools
 - e) Segmentation
